

Cryptography and Network Security

Course Code: **CSEG3040**

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Classical Encryption Techniques

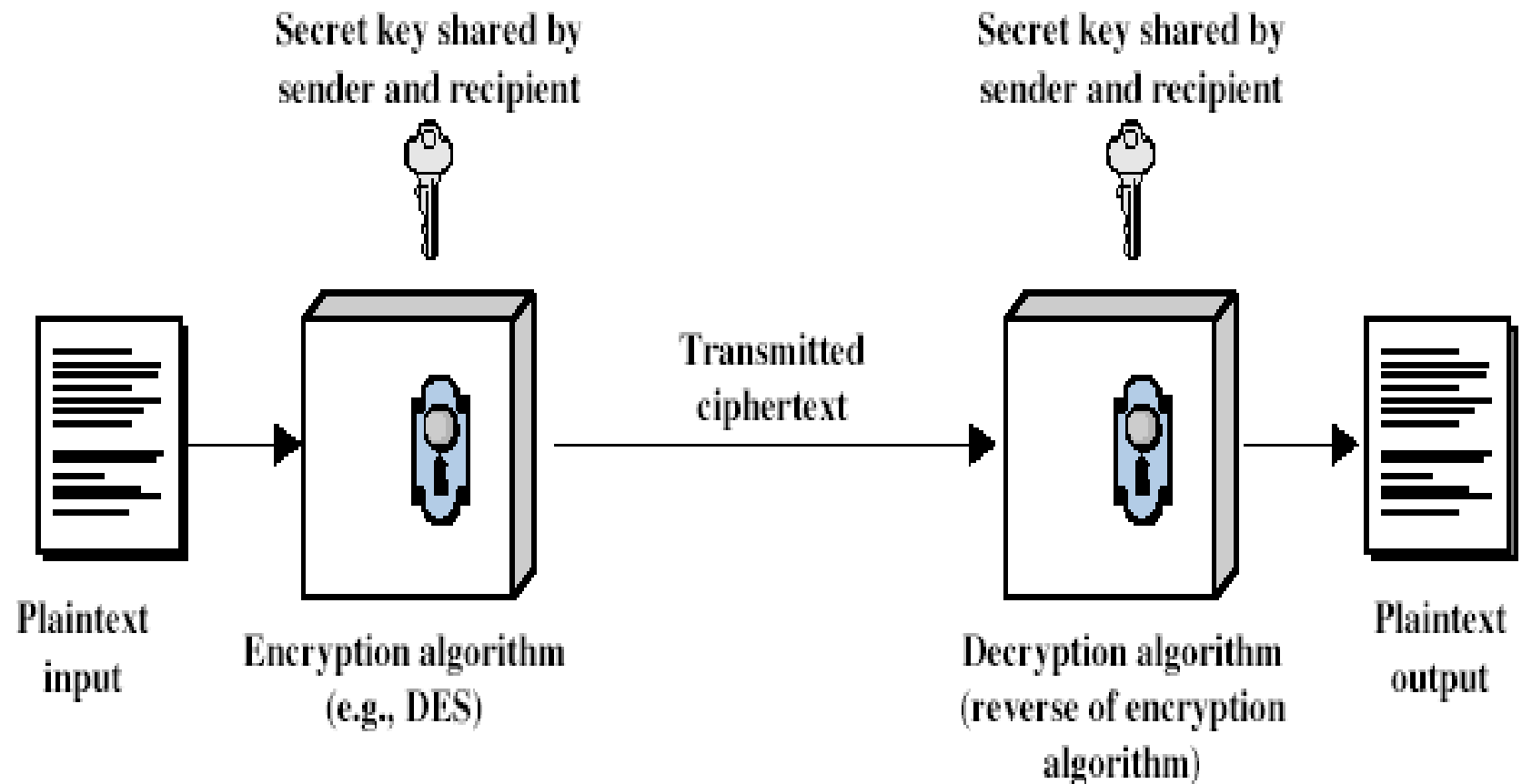
Symmetric Encryption

- or conventional / private-key / single-key
- Sender and recipient share a common key
- All classical encryption algorithms are private-key
- Was the only type prior to invention of public-key in 1970's

Basic Terminology

- **plaintext** - the original message
- **ciphertext** - the coded message
- **cipher** - algorithm for transforming plaintext to ciphertext
- **key** - info used in cipher known only to sender/receiver
- **encipher (encrypt)** - converting plaintext to ciphertext
- **decipher (decrypt)** - recovering ciphertext from plaintext
- **cryptography** - study of encryption principles/methods
- **cryptanalysis (codebreaking)** - the study of principles/methods of breaking ciphertext *without* knowing the key
- **cryptology** - the field of both cryptography and cryptanalysis

Symmetric Cipher Model



Requirements

- two requirements for secure use of symmetric encryption:
 - a strong encryption algorithm
 - a secret key known only to sender / receiver
- $$Y = E_K(X)$$
- $$X = D_K(Y)$$
- assume encryption algorithm is known
 - implies a secure channel to distribute key

Cryptography

- can characterize by:
 - type of encryption operations used
 - substitution / transposition / product
 - number of keys used
 - single-key or private / two-key or public
 - way in which plaintext is processed
 - block / stream

Types of **Cryptanalytic** Attacks

- **ciphertext only**
 - only know algorithm / ciphertext, statistical, can identify plaintext
- **known plaintext**
 - know/suspect plaintext & ciphertext to attack cipher
- **chosen plaintext**
 - select plaintext and obtain ciphertext to attack cipher
- **chosen ciphertext**
 - select ciphertext and obtain plaintext to attack cipher
- **chosen text**
 - select either plaintext or ciphertext to en/decrypt to attack cipher

Brute Force Search

- always possible to simply try every key
- most basic attack, proportional to key size
- assume either know / recognise plaintext

Table 3. Decryption time using Brute force attack method.

Key Length (in bits)	Time to Decrypt
10	Less than 1 second
20	21 seconds
30	6 hours
40	255 days
64	Almost 12,000 years
128	Over 200 septillion years (a number with 27 digits), longer than the life of the universe

A Modified High Capacity Image Steganography Technique Based on Wavelet Transform

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More Definitions

- **unconditional security**
 - no matter how much computer power is available, the cipher cannot be broken since the ciphertext provides insufficient information to uniquely determine the corresponding plaintext
- **computational security**
 - given limited computing resources (eg time needed for calculations is greater than age of universe), the cipher cannot be broken

Classical Substitution Ciphers

- where letters of plaintext are replaced by other letters or by numbers or symbols
- or if plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns

Caesar Cipher

- Earliest known substitution cipher is by Julius Caesar
- First used in military affairs
- Replaces each letter by 3rd letter down
- example:

meet me after the toga party

PHHW PH DIWHU WKH WRJD SDUWB

Caesar Cipher

- can define transformation as:

a b c d e f g h i j k l m n o p q r s t u v w x y z
D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

- mathematically give each letter a number

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12

n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

- then have Caesar cipher as:

$$C = E(p) = (p + k) \bmod (26)$$

$$p = D(C) = (C - k) \bmod (26)$$

Caesar Cipher

- can define transformation as:

a b c d e f g h i j k l m n o p q r s t u v w x y z
D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

- mathematically give each letter a number

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12

n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

- then have Caesar cipher as:

$$C = E(p) = (p + k) \bmod (26)$$

$$p = D(C) = (C - k) \bmod (26)$$

Cryptanalysis of Caesar Cipher

- only have 26 possible ciphers
 - A maps to A,B,..Z
- could simply try each in turn
- a **brute force search**
- given ciphertext, just try all shifts of letters
- do need to recognize when have plaintext
- eg. break ciphertext "GCUA VQ DTGCM"

Cryptanalysis of Caesar Cipher

KEY	PHHW	PH	DIWHU	WKH	WRJD	SDUWB
1	oggv	og	chvgt	vjg	vqic	rctva
2	nffu	nf	bgufs	uif	uphb	qbsuz
3	meet	me	after	the	toga	party
4	ldds	ld	zesdq	sgd	snfz	ozqsx
5	kccr	kc	ydrpc	rfe	rmey	nyprw
6	jbbq	jb	xcqbo	qeb	qldx	mxoqv
7	iaap	ia	wbpan	pda	pkcw	lwnpu
8	hzzo	hz	vaozm	ocz	ojbv	kvmot
9	gyyn	gy	uznyl	nby	niau	julns
10	fxxm	fx	tymxk	max	mhzt	itkmr
11	ewwl	ew	sxlwj	lzw	lgys	hsjll
12	dvvk	dv	rwkvi	kyv	kfxr	grikp
13	cuuj	cu	qvjuh	jxu	jewq	fqhjo
14	btti	bt	puitg	iwt	idvp	epgin
15	assh	as	othsf	hvs	hcuo	dofhm
16	zrrg	zr	nsgre	gur	gbtn	cnegl
17	yqqf	yq	mrfqd	ftq	fasm	bmdfk
18	xppe	xp	lqepc	esp	ezrl	alcej
19	wood	wo	kpdob	dro	dyqk	zkbdi
20	vnnc	vn	jocna	cqn	cxpj	yjach
21	ummb	um	inbmz	bpm	bwoi	xizbg
22	tlla	tl	hmaly	aol	avnh	whyaf
23	skkz	sk	glzcx	znk	zumg	vgxze
24	rjjy	rj	fkyjw	ymj	ytlf	ufwyd
25	qiix	qi	ejxiv	xli	xske	tevxz

Figure 3.3 Brute-Force Cryptanalysis of Caesar Cipher

Monoalphabetic Cipher

- rather than just shifting the alphabet
- **could shuffle (jumble) the letters arbitrarily**
- each plaintext letter maps to a different random ciphertext letter
- hence key is 26 letters long

Plain: abcdefghijklmnopqrstuvwxyz

Cipher: DKVQFIBJWPESCXHTMYAUOLRGZN

Plaintext: ifwewishtoreplaceletters

Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA

One interesting Key is: the quick brown fox jumps over the lazy dog

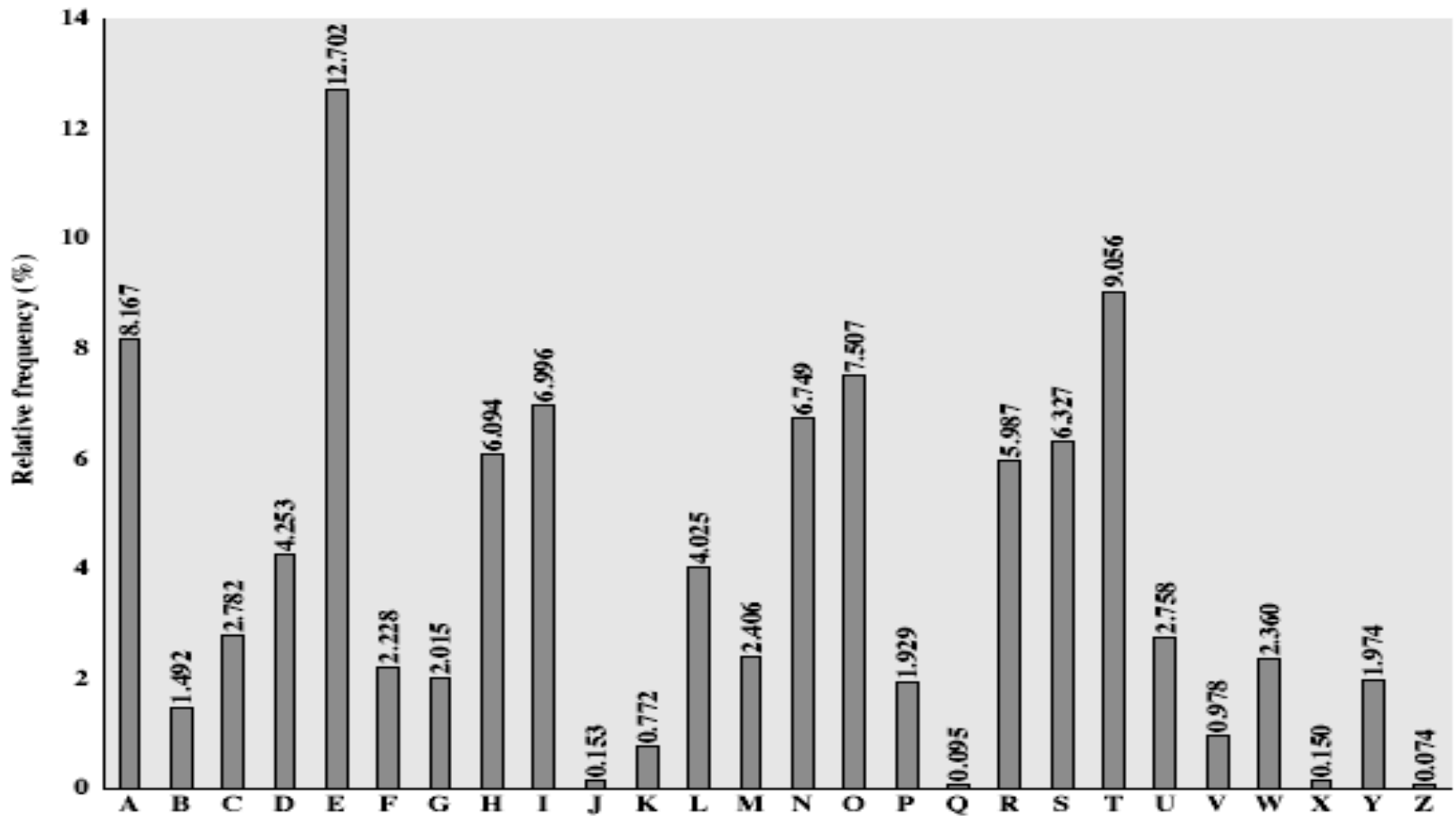
Monoalphabetic Cipher Security

- now have a total of $26!$ ($= 4 \times 10^{26}$) keys
- with so many keys, might think is secure but would be **!!!WRONG!!!**
- Problem is **language characteristics**

Language Redundancy and Cryptanalysis

- human languages are **redundant**
 - eg "th lrd s m shphrd shll nt wnt"
- letters are not equally commonly used
- in English **e is by far the most common letter**
then T,R,N,I,O,A,S
- **other letters are fairly rare** such as Z,J,K,Q,X
- have tables of letter frequencies for:
 - single,
 - double
 - triple

English Letter Frequencies



Use in Cryptanalysis

- key concept - monoalphabetic substitution ciphers do not change relative letter frequencies
 - discovered by Arabian scientists in 9th century
 - calculate letter frequencies for ciphertext
- compare counts/plots against known values
- for monoalphabetic must identify each letter
 - tables of common double/triple letters help

Example Cryptanalysis

- Given ciphertext:

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

- Count relative letter frequencies (see text)
- Guess **P & Z** are **e and t**
- Guess “**ZW** is **th**” and hence “**ZWP** is **the**”
- Proceeding with trial and error finally we get:
*it was disclosed yesterday that several informal but
direct contacts have been made with political
representatives of the viet cong in moscow*

Polyalphabetic Ciphers

- Another approach to improving security is to use multiple cipher alphabets
- called **polyalphabetic substitution ciphers**
- Makes cryptanalysis harder with more alphabets to guess and flatter frequency distribution
- **use a key to select which alphabet is used for each letter of the message**
- use each alphabet in turn
- repeat from start after end of key is reached

Vigenère Cipher

- simplest polyalphabetic substitution cipher is the **Vigenère Cipher**
- effectively multiple caesar ciphers
- key is multiple letters long $K = k_1 k_2 \dots k_d$
- i^{th} letter specifies i^{th} alphabet to use
- use each alphabet in turn
- repeat from start after d letters in message
- decryption simply works in reverse

Example

- write the plaintext out
- write the keyword repeated above it
- use each key letter as a caesar cipher key
- encrypt the corresponding plaintext letter
- eg using keyword ***deceptive***

key: **deceptive**deceptivedeceptive

key-value: 3424...etc

plaintext: wearediscoveredsaveyourself

ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

<https://cryptii.com/>

Chosen plaintext can easily break the key. (Try entering plaintext as
aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa)

Playfair Cipher

- Best-known multiple-letter encryption cipher
- Treats digrams in the plaintext as single units
- Based on 5 x 5 matrix of letters constructed using a key

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Playfair Cipher

Preprocess Plaintext

Remove duplicate letters in keyword before filling matrix.

Replace repeated letters in a digraph with filler X.

Example: BALLOON → BA LX LO ON

Encryption Rules

Same Row → replace each letter with one to its right (wrap around).

Example: AR → RM

Same Column → replace each letter with one below (wrap around).

Example: MU → CM

Rectangle Rule → each letter replaced by one in its row, column of the other letter.

Example: HS → BP, EA → IM (or JM).

Hill Cipher

THE HILL ALGORITHM This encryption algorithm takes m successive plaintext letters and substitutes for them m ciphertext letters. The substitution is determined by m linear equations in which each character is assigned a numerical value ($a = 0, b = 1, \dots, z = 25$). For $m = 3$, the system can be described as

$$c_1 = (k_{11}p_1 + k_{21}p_2 + k_{31}p_3) \bmod 26$$

$$c_2 = (k_{12}p_1 + k_{22}p_2 + k_{32}p_3) \bmod 26$$

$$c_3 = (k_{13}p_1 + k_{23}p_2 + k_{33}p_3) \bmod 26$$

This can be expressed in terms of row vectors and matrices:⁶

$$(c_1 \ c_2 \ c_3) = (p_1 \ p_2 \ p_3) \begin{pmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{pmatrix} \bmod 26$$

or

$$\mathbf{C} = \mathbf{PK} \bmod 26$$

One-Time Pad

- if a truly random key as long as the message is used, the cipher will be secure
- called a One-Time pad
- is unbreakable since ciphertext bears no statistical relationship to the plaintext
- since for **any plaintext** & **any ciphertext** there exists a key mapping one to other
- **can only use the key once** though
- have problem of **safe distribution** of key

One-Time Pad

ANKYODKYUREPFJBYOJDSPLREYIUNOFDOIUERFPLUYTS

We now show two different decryptions using two different keys:

ciphertext: ANKYODKYUREPFJBYOJDSPLREYIUNOFDOIUERFPLUYTS
key: *pxlmvmsydofuyrvzwc tnlebnecvgdupahfzzlmnyih*
plaintext: mr mustard with the candlestick in the hall

ciphertext: ANKYODKYUREPFJBYOJDSPLREYIUNOFDOIUERFPLUYTS
key: *pftgpmiydgaxgoufhklllmhsqdqogtewbqfgyovuhwt*
plaintext: miss scarlet with the knife in the library

Transposition Ciphers

- Now consider classical **transposition** or **permutation** ciphers
- These hide the message by rearranging the letter order
- Without altering the actual letters used
- Can recognise these since have the same frequency distribution as the original text

Rail fence technique

Depth: 3

m				m				t				h				r		
	e		t		e		f		e		t		e		a		t	
		e				a				r				p				y

Plaintext: *meet me after the party*

Ciphertext: MMTHRETEFETEATEARPY

Too simple, if the plaintext is too short or the length of the plaintext is equal to the depth it reveals the plaintext.

Row Transposition Ciphers

- a more complex scheme
- **write** letters of **message** out in **rows** over a **specified number of columns**
- **then reorder the columns** according to **some key** before reading off the rows

Key: 4 3 1 2 5 6 7

Plaintext: a t t a c k p

o s t p o n e

d u n t i l t

w o a m x y z

Ciphertext: TTNAAPTMTSU0AODWCOIXKNLYPETZ

Note: to add complexity, use **multi-stage** transposition. Input the ciphertext in the above process as plaintext to generate the next stage of ciphertext.

Product Ciphers

- Ciphers using substitutions or transpositions are **not secure because of language characteristics**
- Hence, consider using several ciphers in succession to make harder, but:
 - two substitutions make a more complex substitution
 - two transpositions make more complex transposition
 - but a substitution followed by a transposition makes a new much harder cipher
- This is bridge from classical to modern ciphers

Rotor Machines

before modern ciphers, rotor machines were most common cipher using **multi-stage**.

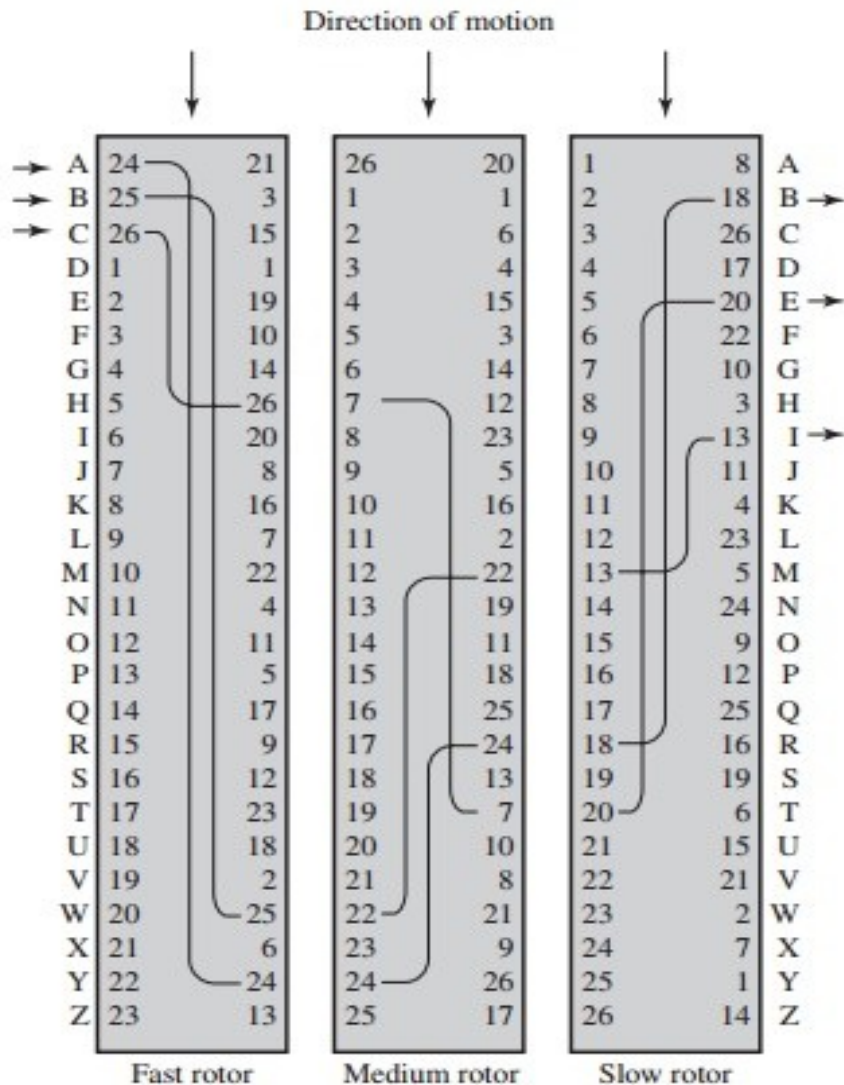
- were widely used in World War-II
 - German Enigma, Allied Hagelin, Japanese Purple
- implemented a very complex, varying substitution cipher
- used a series of cylinders, each giving one substitution, which rotated and changed after each letter was encrypted
- with 3 cylinders have $26^3=17576$ alphabets

Rotor Machines

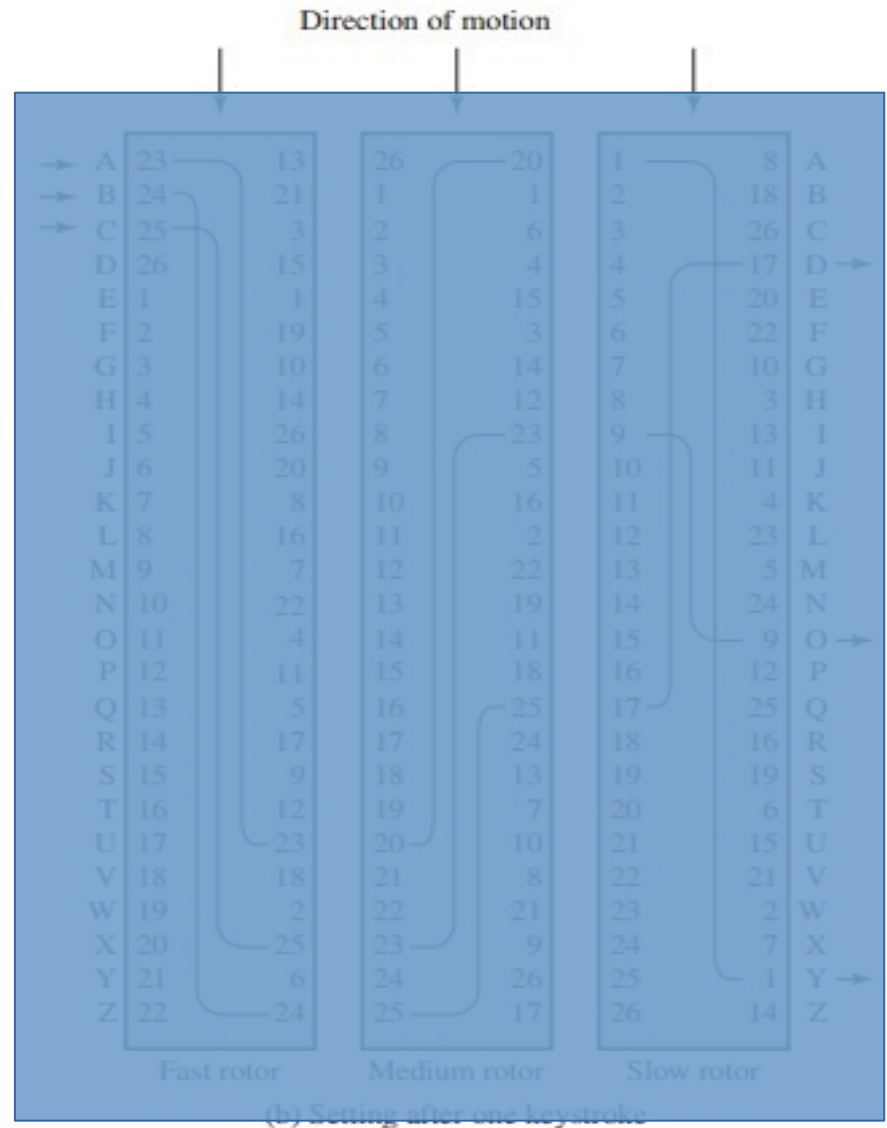


Rotor Machines

A B C --> B I E



(a) Initial setting



(b) Setting after one keystroke

Figure 2.8 Three-Rotor Machine with Wiring Represented by Numbered Contacts

Rotor Machines

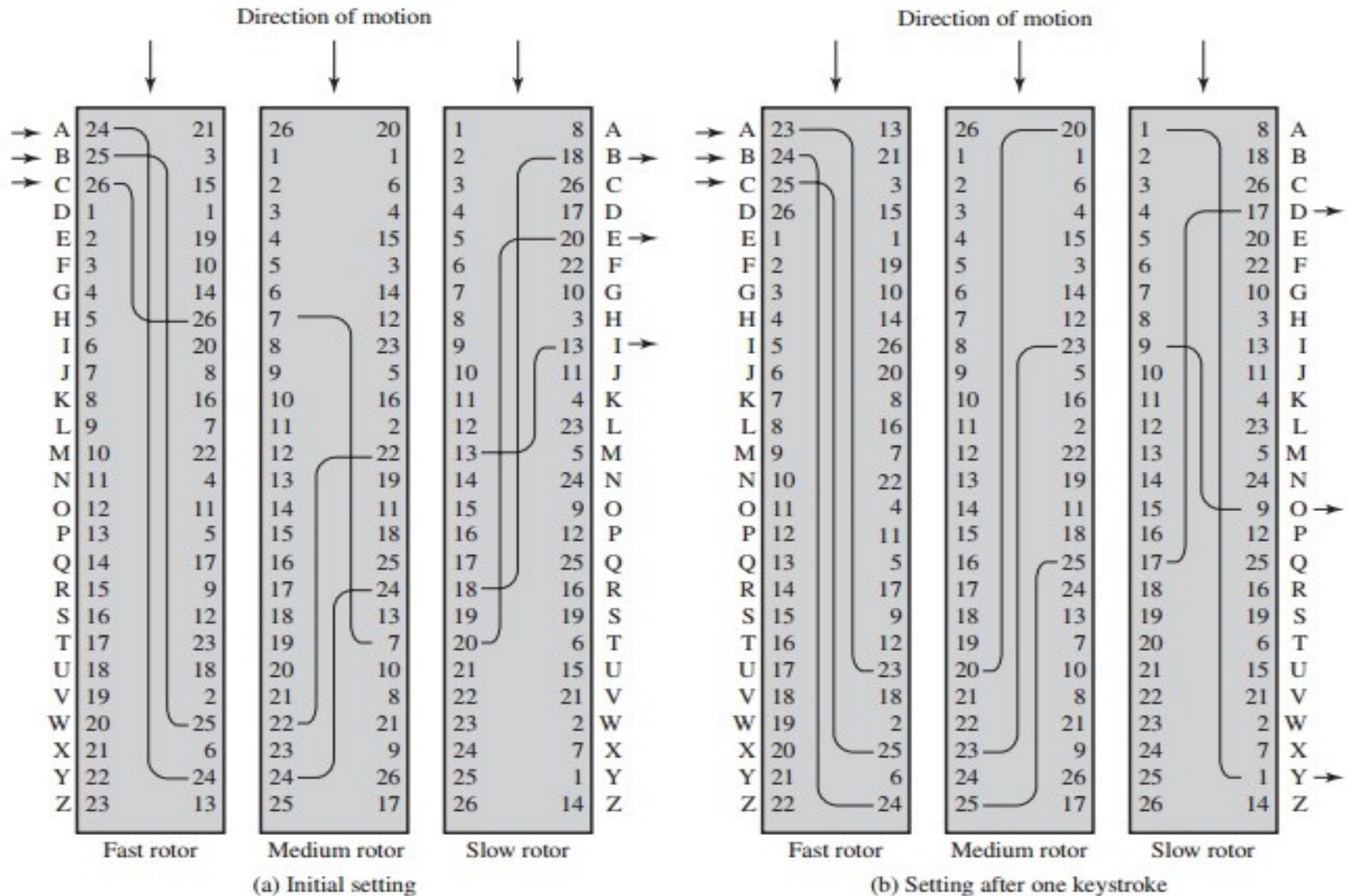


Figure 2.8 Three-Rotor Machine with Wiring Represented by Numbered Contacts

Steganography

*Meeting you felt like stepping into a beautiful **dream**.*

*Every moment with you fills my life with warmth and good **company**.*

*Nothing brightens my mornings quite like watching the sunrise with you at **dawn**.*

*All the little things you do wrap around me in a comforting **aura**.*

*My memory of your laugh stays with me like a cherished **poem**.*

*Even when the world is noisy, your quiet voice brings a sincere **smile**.*

*In your presence, the simplest dinner becomes something special — sometimes even **spaghetti**.*

*Some evenings we share are so perfect I find myself holding them in my mind **always**.*

*And when I plan for tomorrow, every hopeful thought starts with a small, bright **idea**.*

*Moments apart only deepen my longing; returning to you brings a steady **calm**.*

*I imagine our future filled with laughter, festivals, and you in a beautiful **sari**.*

*Through every storm and joy, I promise to stand by you with all my **heart**.*

Steganography

- An alternative to encryption
- Hides existence of message
 - using only a subset of letters/words in a longer message marked in some way
 - using invisible ink
 - hiding in LSB in graphic image or sound file
- Has drawbacks
 - high overhead to hide relatively few info bits

Summary

- have considered:
 - classical cipher techniques and terminology
 - monoalphabetic substitution ciphers
 - cryptanalysis using letter frequencies
 - Playfair ciphers
 - polyalphabetic ciphers
 - transposition ciphers
 - rotor machines
 - Stenography
- Use **<https://cryptii.com/>**